

Supplementary Material: Lane Perception for Automated Driving with Human-Anchored, Engine-Certified Ground Truth

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Abstract

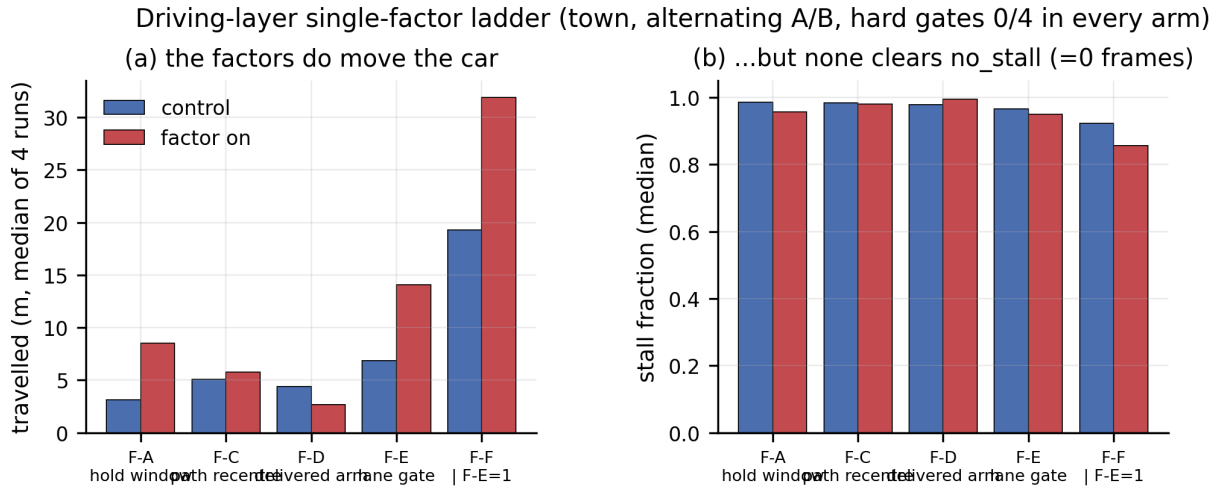
This supplement carries the sixteen figures whose readings are quoted in the main text but which are not load-bearing for its argument. Every figure is regenerated from the same recorded artefacts as the main text (per-run scorecards, per-frame telemetry traces, run manifests, gate verdict files and sealed-set records) by `m5_paper_figures.py`, `m5_paper_figures_ext.py` and `m5_paper_figures_extra.py`; no number in this supplement is transcribed by hand. Section S-A collects the driving-layer evidence behind Section V-E of the main text (the single-factor ladder, per-run traces, arm distributions, lane-placement deviation, safety margins and the pairing comparison); S-B collects the composition and post-processing scans behind Section V-C (dose response over five metrics, parallel-merge and threshold scans, SWA, the $\beta=0.8$ loss arm, instance coverage and the negative-side family); S-C carries the training history; S-D carries the cost ledger and the spatial-isolation audit. The reading conventions that apply to all of them are stated in the last section, including the three quantities that are UNKNOWN rather than zero.

Index Terms—Supplementary figures, negative results, single-factor ablation, driving acceptance,

S-A. Driving-layer evidence

The main text reports five pre-registered driving factors, of which four were implemented and rejected and one was refuted

by prior measurement before implementation, and localises the blocker to a static body-sweep deadlock. The figures below are the per-arm and per-frame evidence behind those statements.



worst-case body-centre-cross frames (control→factor): 29→30 / 84→39 / 36→49 / 23→66 / 23→31

Fig. S1. Driving single-factor ladder: travelled distance (a) and stall fraction (b) per arm, with worst-case body-centre-cross frames.

Every arm in S1 still fails the `no_stall` hard target (0 frames) and the hard gates stay 0/4; the ladder is what justifies

reporting the implemented factors as rejected rather than as partial progress.

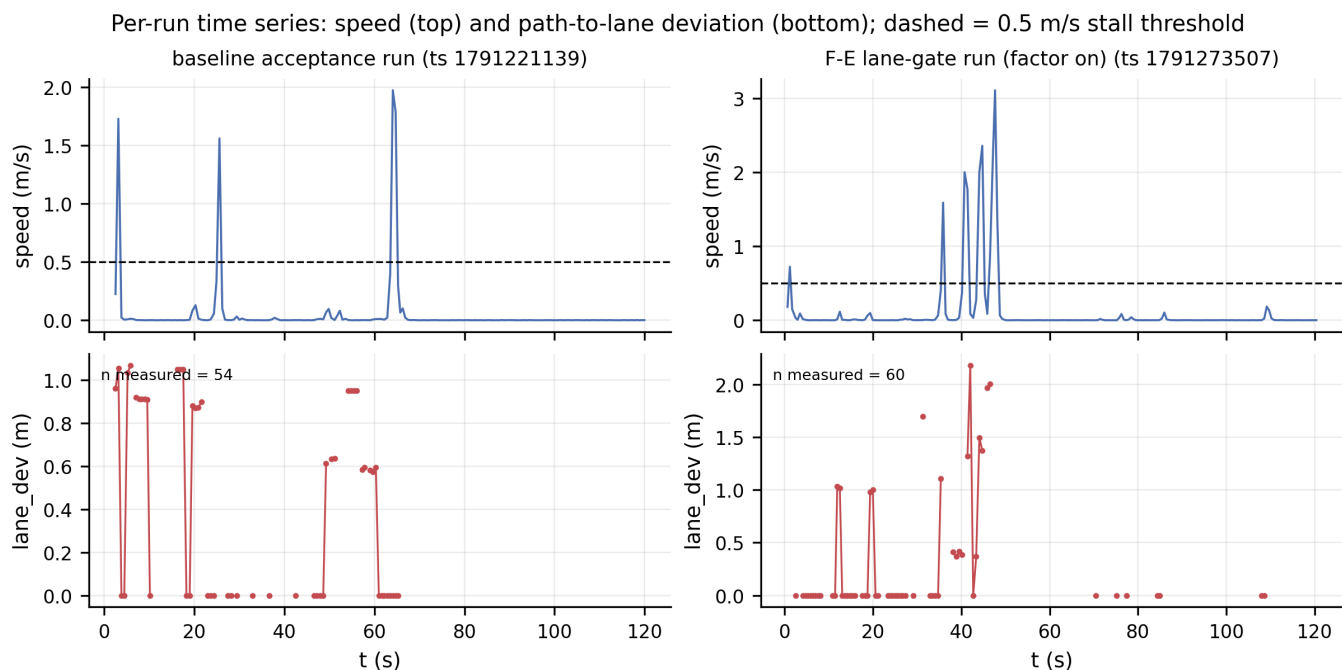


Fig. S24. Per-run time series (speed top, path-to-lane deviation bottom) for a baseline acceptance run and an F-E lane-gate run.

S24 shows why the stall is an availability failure and not a control failure: speed stays at zero for long plateaus while the

deviation series shows the car parked off the lane centre, and the F-E arm moves further only by travelling with a larger deviation.

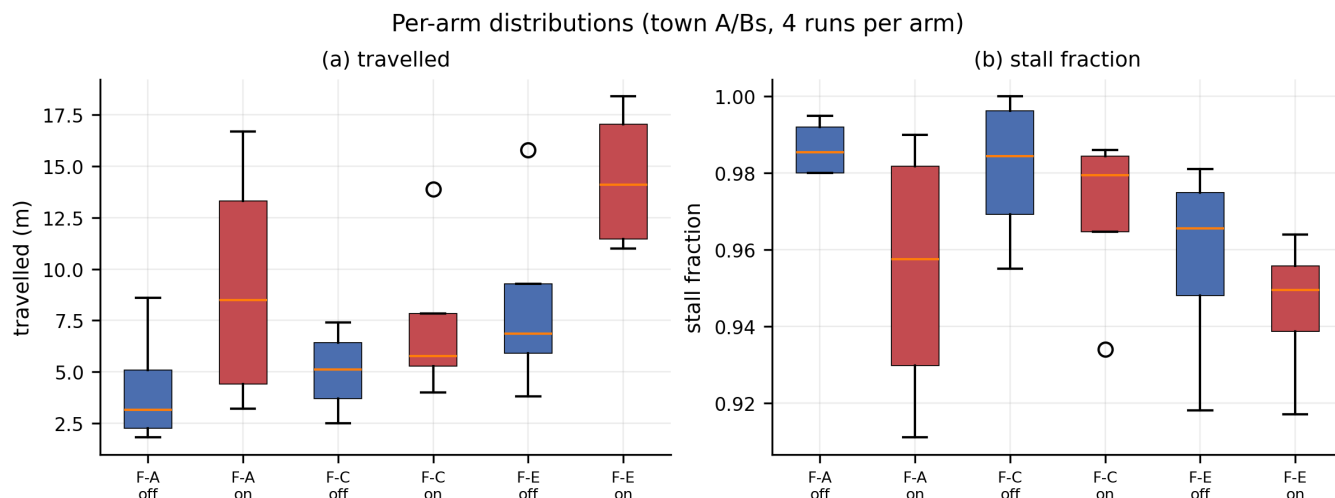


Fig. S25. Per-arm distributions (travelled, stall fraction) for the F-A/F-C/F-E A/Bs.

The arm-level distributions in S25 are the raw material of the pre-registered maximum rule: verdicts were read as the worst

run per arm, and the overlap of the ranges is why the delivered perception arm produced no separation on path availability (0.179 $\rightarrow$ 0.200).

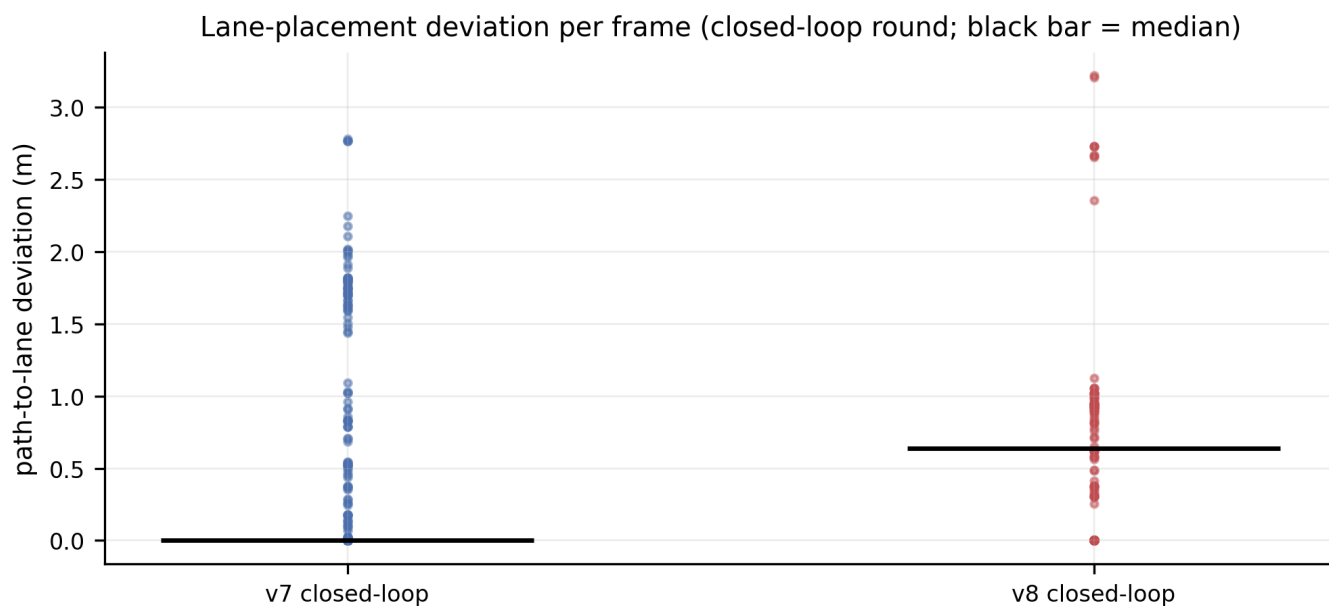


Fig. S26. Per-frame lane-placement deviation in the closed-loop round (v7 vs v8; black bar = median).

S26 is the closed-loop counterpart of the definitional result: under the baseline definition the median lane deviation is 0.86

m with the body crossing the centre line in 8/8/8/13 frames, and under the adopted definition the crossings vanish at the cost of availability.

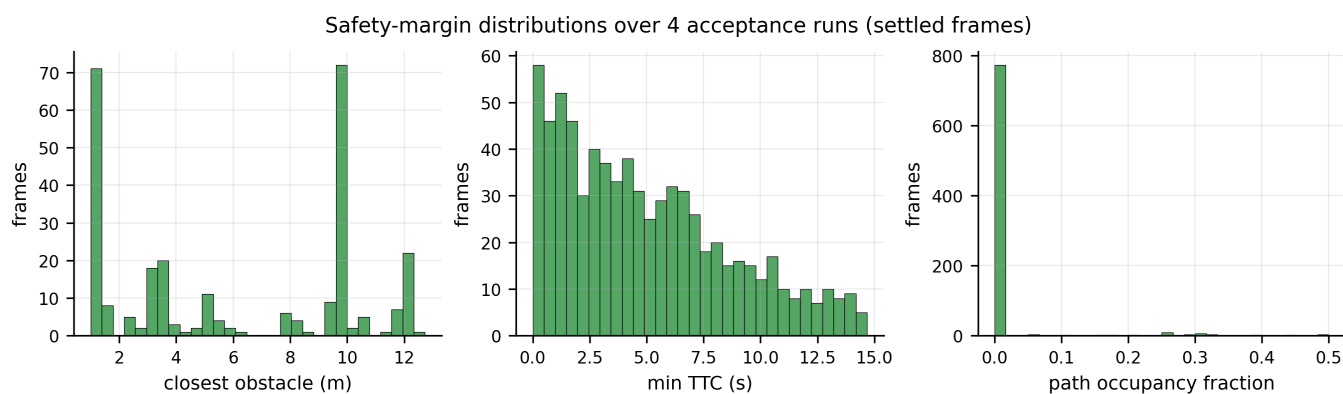


Fig. S27. Safety-margin distributions (closest obstacle, minimum time-to-collision, path occupancy) over four acceptance runs.

The margins in S27 are what backs the statement that the acceptance failures are availability failures, not safety failures:

no collision, no off-road frame and no reversing frame occurred in any of the four runs, and the measured margins stay inside their distributions' support rather than at their limits.

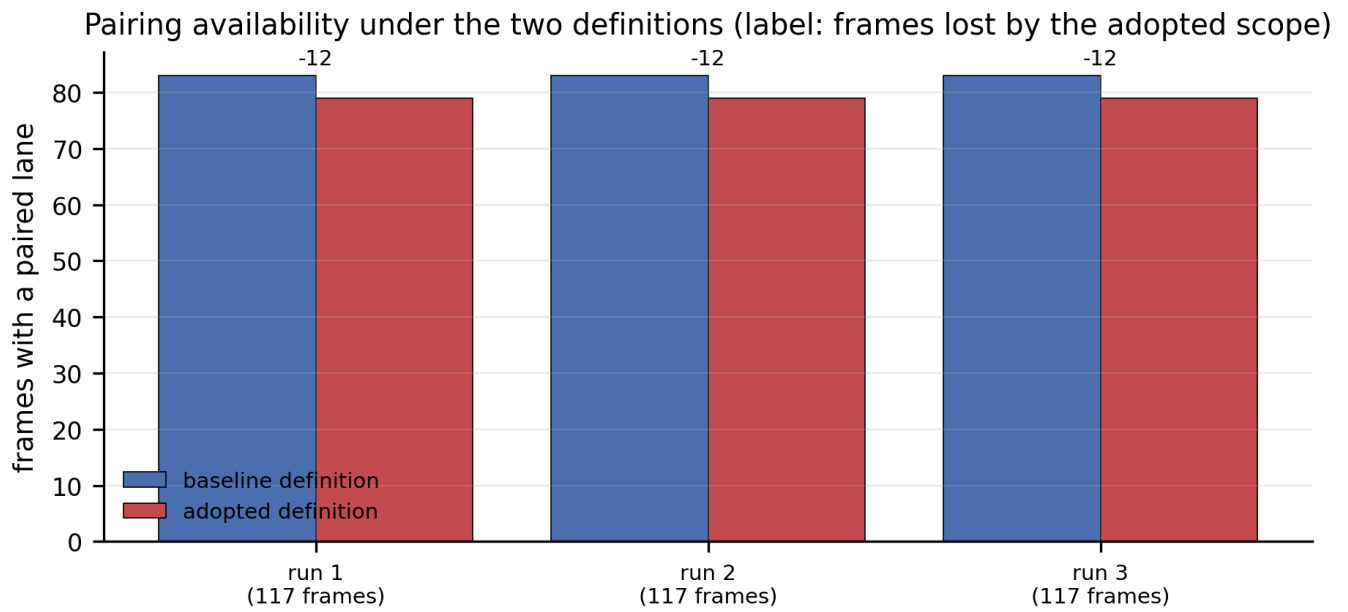


Fig. S37. Pairing comparison: paired-frame and candidate counts per arm (dev and limited-class pools).

S37 quantifies the availability cost of the adopted definition at the pairing stage itself: 3–4 paired frames are lost per 117-frame run, which is the mechanism behind the closed-loop availability drop reported in the main text.

S-B. Composition and post-processing scans

These are the scans behind the adopted composition (negative dose 6×) and the adopted protocol (lateral scope 5.5 m, merge radius 1.0 m, appearance gate on).

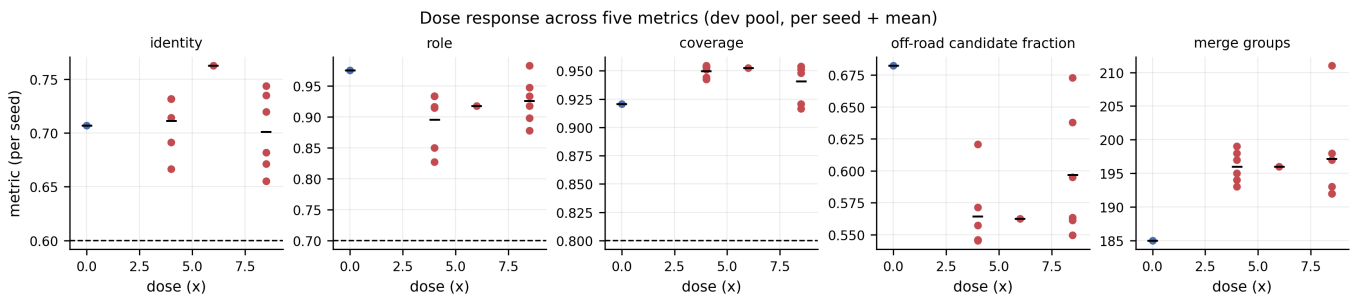


Fig. S7. Dose response across five metrics (identity, role, coverage, off-road candidate fraction, merge groups), per seed and mean.

In S7 identity and role move with the negative dose while coverage and the off-road candidate fraction barely do, which

is why the dose was varied as the composition axis instead of a loss-weight knob.

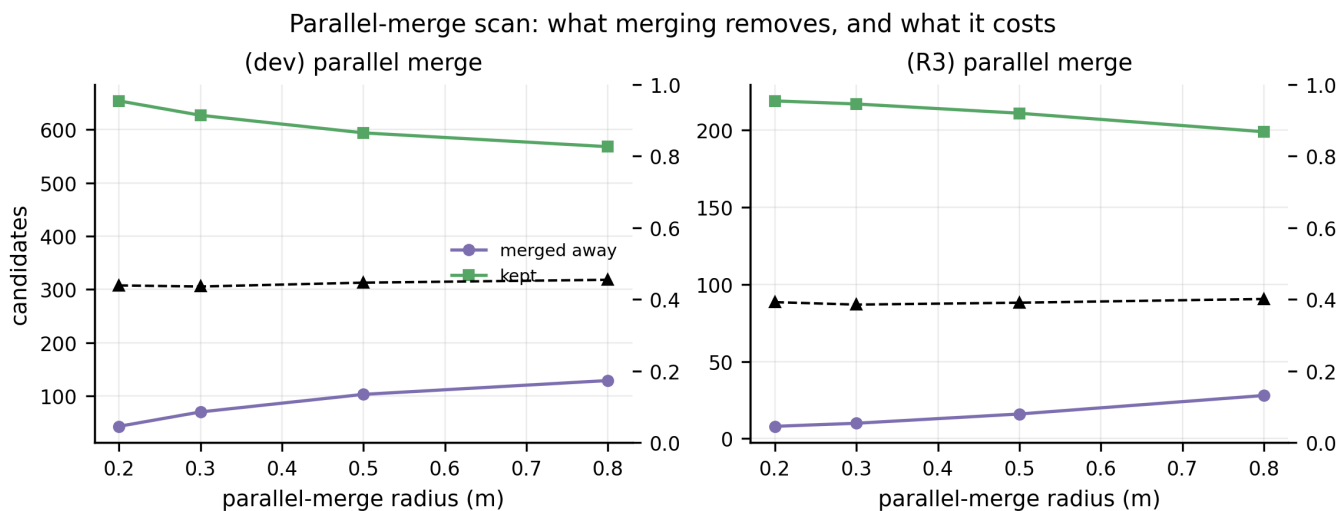


Fig. S8. Parallel-merge scan: candidates merged away and kept, with the identity consequence.

S8 is why the adopted merge radius is 1.0 m rather than the aggressive end of the sweep: tightening the radius removes

near-duplicate same-side candidates (dev 477 → 166 kept from 6.0 m to 3.0 m) but costs identity.

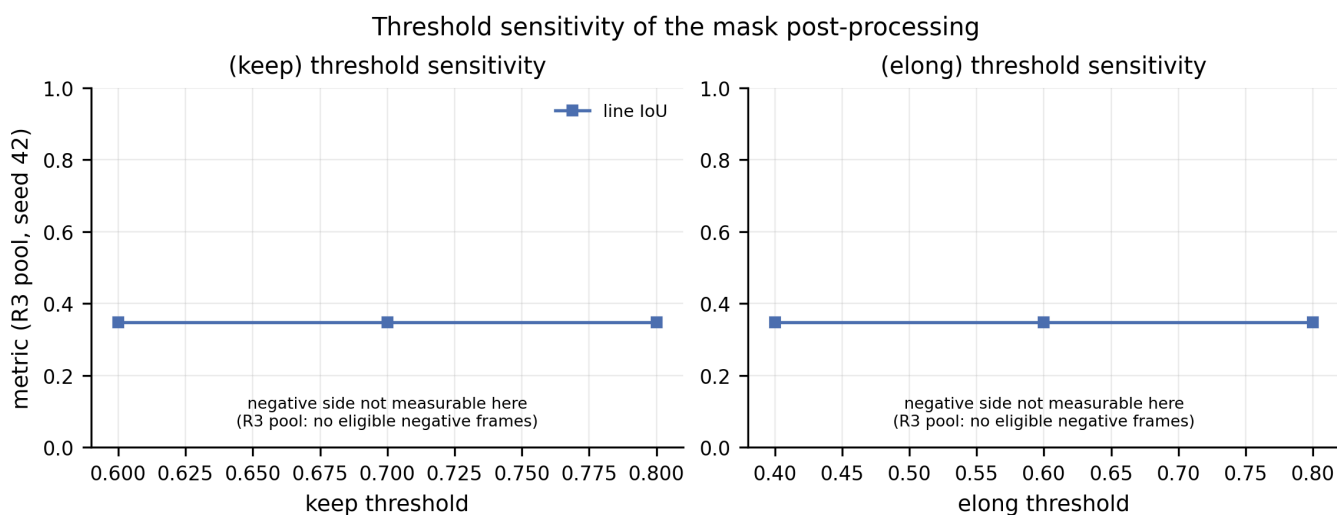


Fig. S9. Threshold sensitivity of the mask post-processing (keep and elongation thresholds).

The limited-class pool carries no eligible negative frames, so the false-positive series in S9 is not measurable there; the gap

is marked in the figure and is not read as a zero.



Fig. S10. SWA single-factor result: no seed is rescued by last-five-epoch averaging.

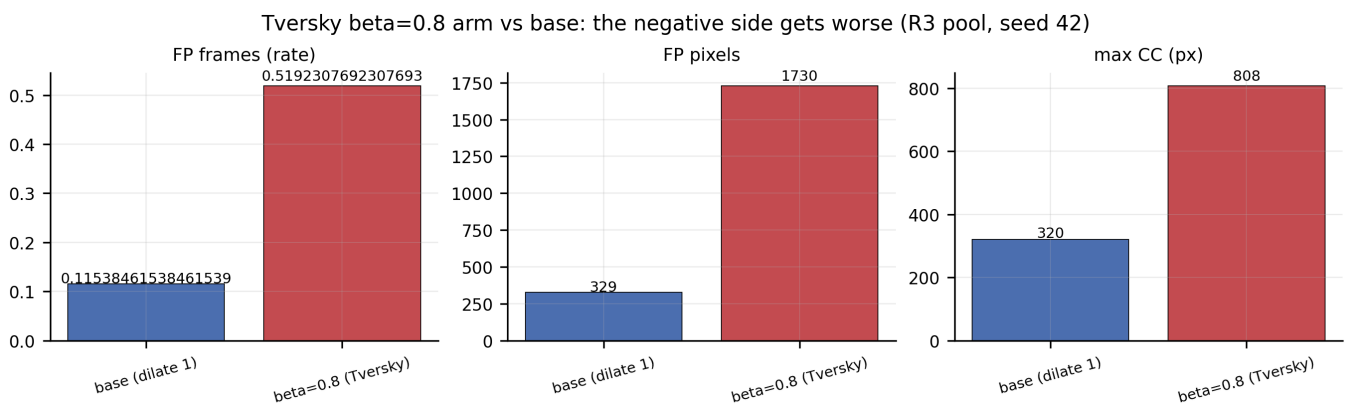


Fig. S17. Tversky $\beta=0.8$ arm versus base on the negative side (false-positive frames, false-positive pixels, largest connected component).

S10 and S17 are the two rejected single factors of Section V-C: weight averaging rescued no seed, and the Tversky loss arm

tripled the false-positive frame rate (11.5% $\rightarrow$ 51.9% of eligible frames; 329 $\rightarrow$ 1730 false-positive pixels; largest component 320 $\rightarrow$ 808 px).

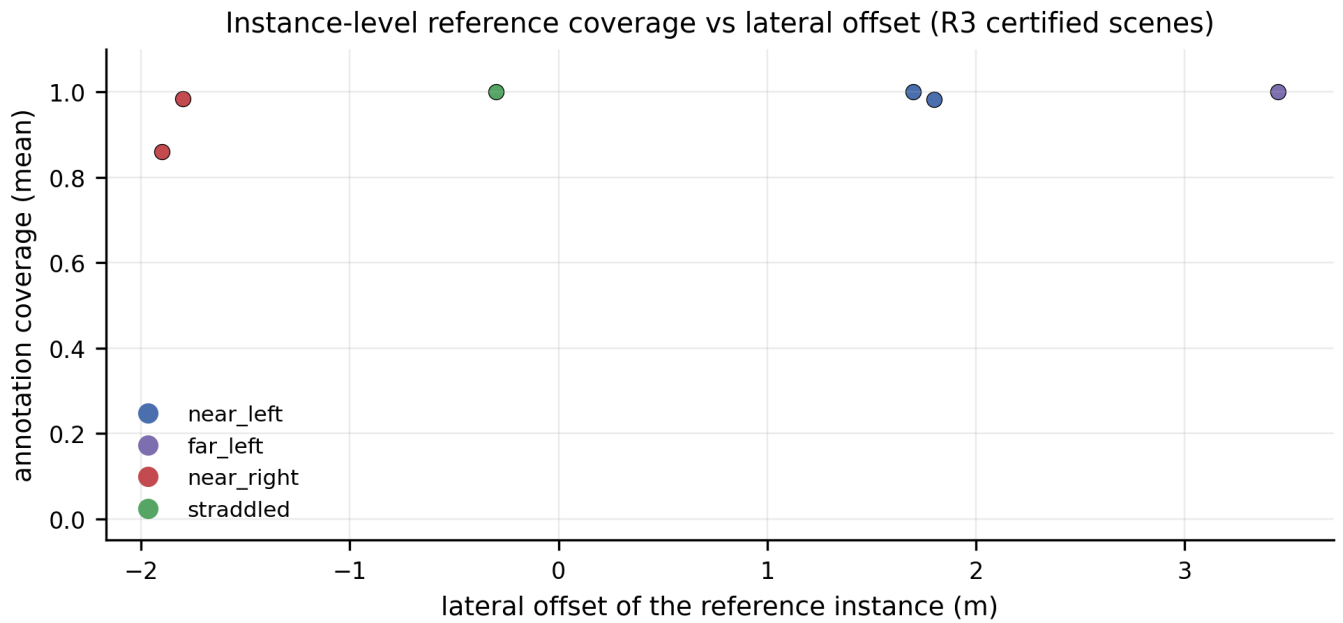


Fig. S19. Instance-level reference coverage versus lateral offset across the certified limited-class scenes, coloured by role.

S19 explains why the appearance gate can afford to be aggressive: across the certified limited-class scenes the

reference instances are covered by annotation at essentially 1.0 regardless of lateral offset, so a surviving candidate is still measured against complete truth.

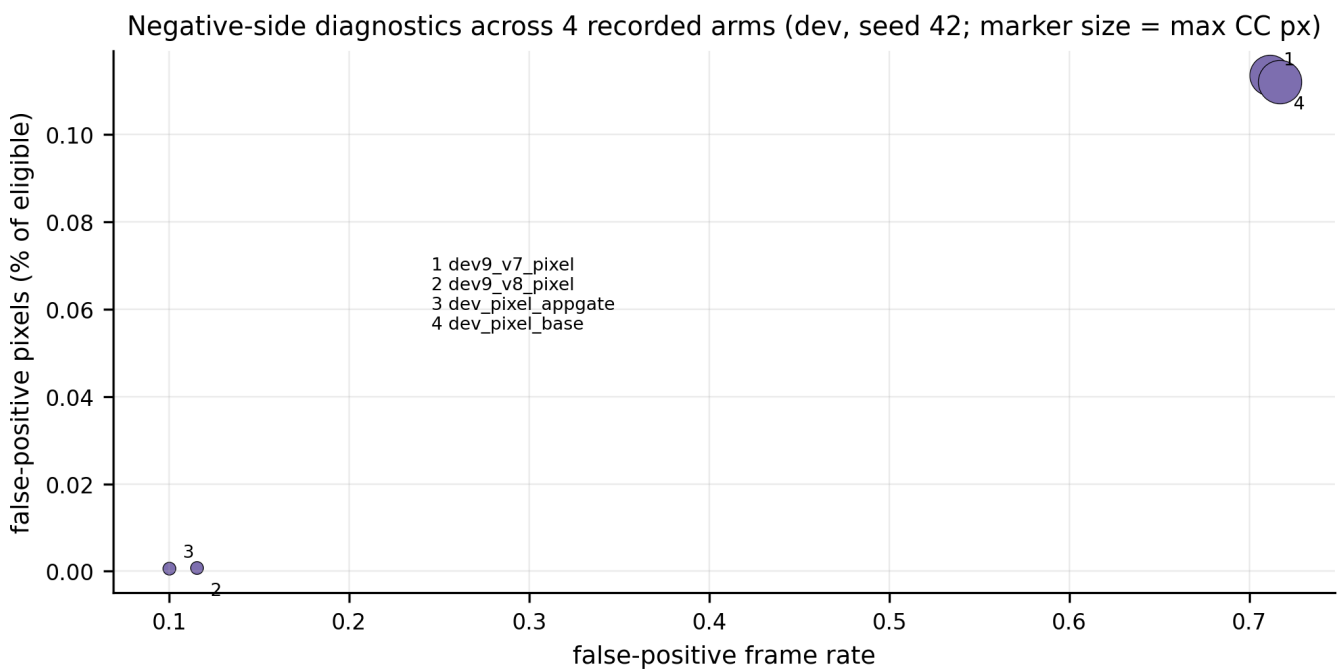


Fig. S20. Negative-side diagnostics across all recorded pixel arms (false-positive frame rate versus false-positive pixel fraction; marker size = largest connected component).

Across every recorded pixel arm the negative-side readings in S20 cluster rather than spread, which is the measurement that

motivated varying the negative-example dose instead of a loss-weight knob.

S-C. Training history

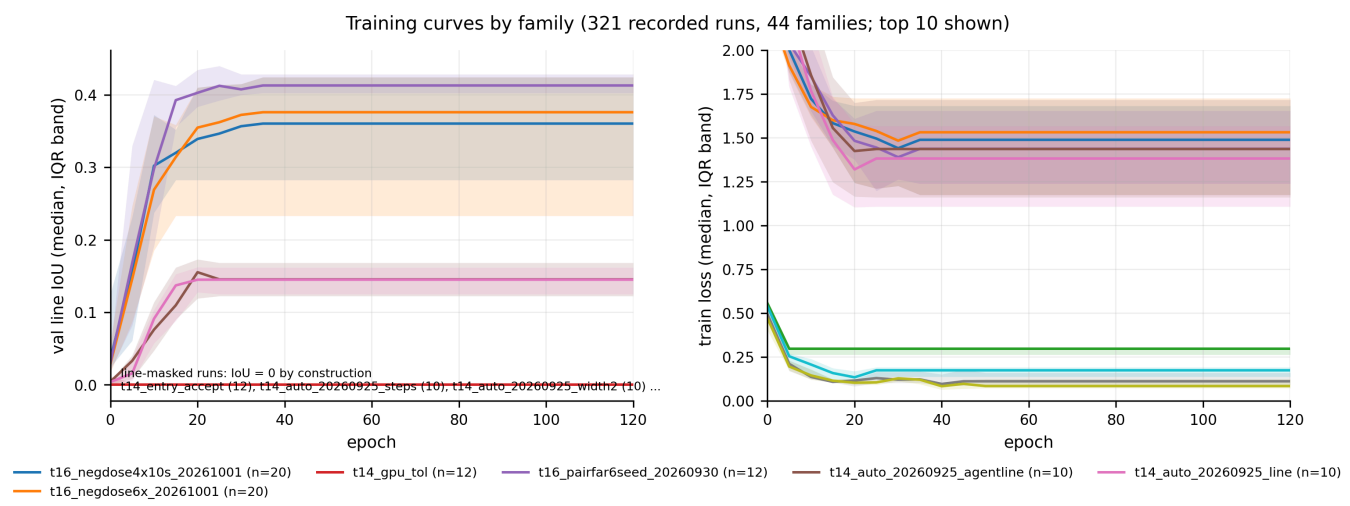


Fig. S21. Training curves from the recorded 'train_hist.json' runs (validation line IoU and training loss per epoch).

The programme's training history comprises 321 recorded runs across 44 families; S21 shows the representative curves. Families whose line channel was masked sit at zero line IoU by construction rather than by failure and are annotated as such in

the figure, so their curves are not negative results. Runs converge within 20–40 epochs of the 120-epoch budget.

S-D. Cost and data-governance audits

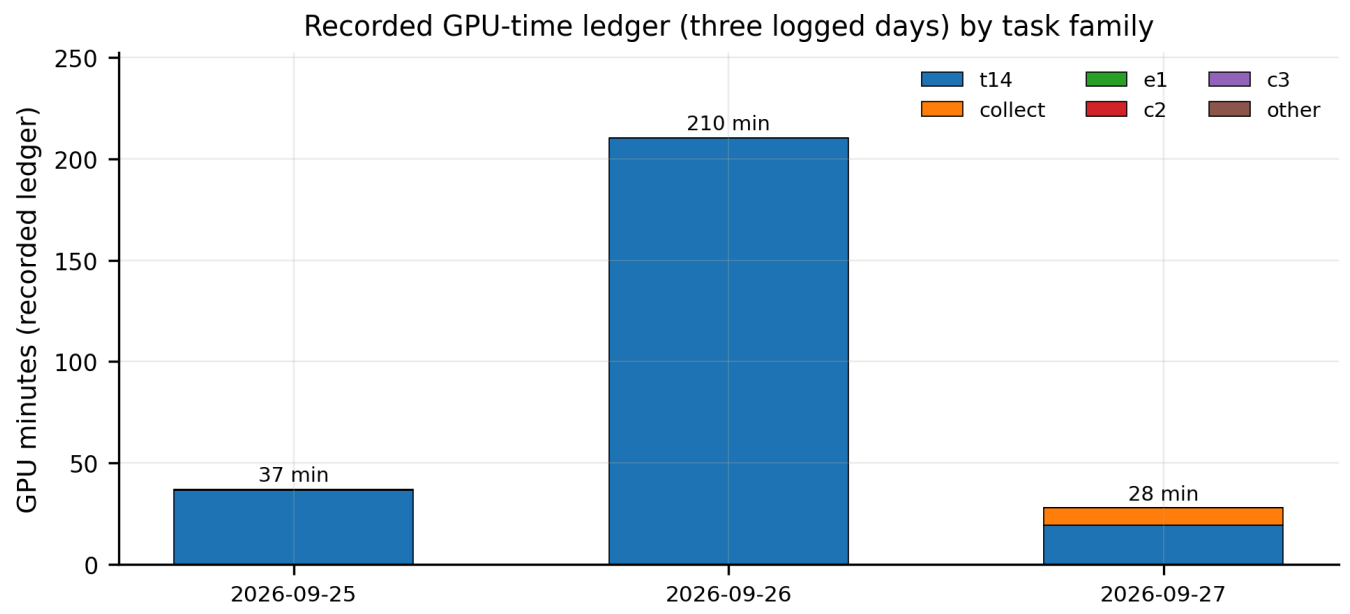


Fig. S35. GPU cost ledger across the programme's recorded rounds.

S35 is the machine-readable ledger kept for every unattended round; it records cost rather than estimating it, and it is the

artefact behind the statement that the reported rounds added no new manual annotation (the labour was machine time).

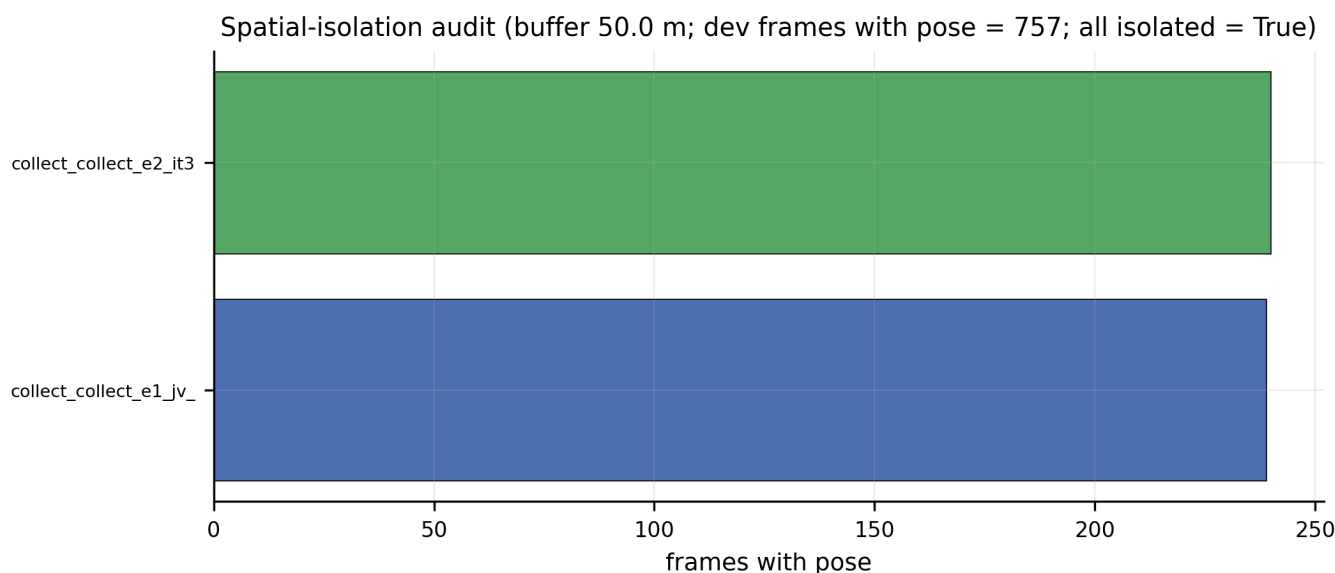


Fig. S41. Spatial-isolation audit: every candidate collection against the development anchors under a 50 m buffer.

S41 is the isolation audit that admitted the reference and sealed sets: disjointness is checked with pose under a 50 m buffer, not with directory names, and each candidate composition gets a recorded verdict. The same machinery rejected the first two compositions of the sealed set.

S-E. Reading conventions and stated gaps

- 1. Unit of judgement.** The acceptance results are six-seed arms (mean read against the gate, every seed reported); the one-shot confirmation is a single checkpoint selected by index, not by score.
- 2. The sealed set is consumed.** The confirmation figures and the main-text confirmation come from that single consumption; the instance-level probes (identity, role, reference coverage) were **not** run on it and are recorded as UNKNOWN, not as passes.
- 3. Two definitions of "line".** The paint scope is used for the adopted verdict, the label scope keeps comparability with the baseline; dilation is a trade-off between the two sets (the 0 px row is a mean read, the 1/2 px rows are strict reads).
- 4. Frame counts are confounded by motion.** `body_cross_centre_frames` scales with how far the car travelled; the F-C and F-E verdicts were taken under the pre-registered maximum-over-runs rule and a per-distance reading was pre-registered separately for later rounds.

- 5. Dose response re-evaluated.** The earlier curve's identity values do not reproduce under the frozen protocol (the same 6× checkpoint reads 0.763 there and 0.667 here), so it was replaced by a paired-seed re-evaluation (seeds 42–47): 0× 0.658, 4× 0.661, 6× 0.665 — indistinguishable — while the recorded 8.5× value (0.701) is higher than 6×. The 8.5× checkpoints were removed in a storage pass, so that arm cannot be re-evaluated.
- 6. No eligible negative frames on the limited-class pool,** so that pool's false-positive rate is not measurable (Figure S15) and is never reported as zero.
- 7. Schematics are not measurements.** The five schematic figures in the main text are drawn from the code and document definitions, not from run data.
- 8. Artefact retention.** Per-epoch snapshots and superseded run directories were removed in a storage pass (audit file `logs/_cleanup_20261006.txt`); every verdict file, scorecard, manifest, sealed frame and the delivered checkpoint are retained, so every figure in the main text and in this supplement is reproducible from the retained artefacts.

S-F. Individual panels of the main-text composites

The seven composite figures in the main text are assembled from the 26 individual figures below, reproduced here at full size so that any panel can be inspected without the composite's scaling. The panel letter in each caption refers to its position in the composite.

Candidate-scope pipeline (schematic) and the measured revocation

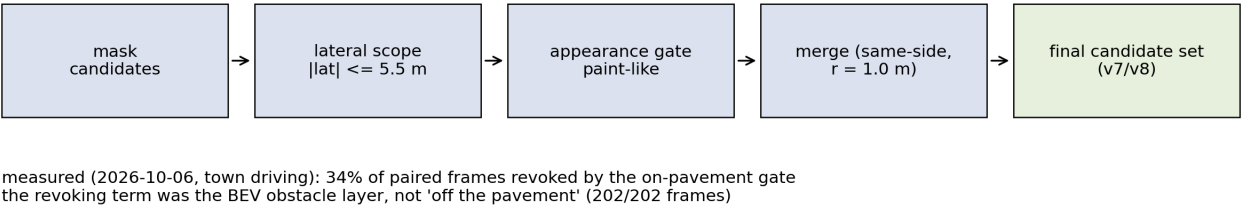


Fig. S17. Panel (a) of the main-text design and contract composite. Candidate-scope pipeline (schematic) with the measured revocation rate.

Counting contract v5 (schematic): every ratio has an explicit numerator and denominator

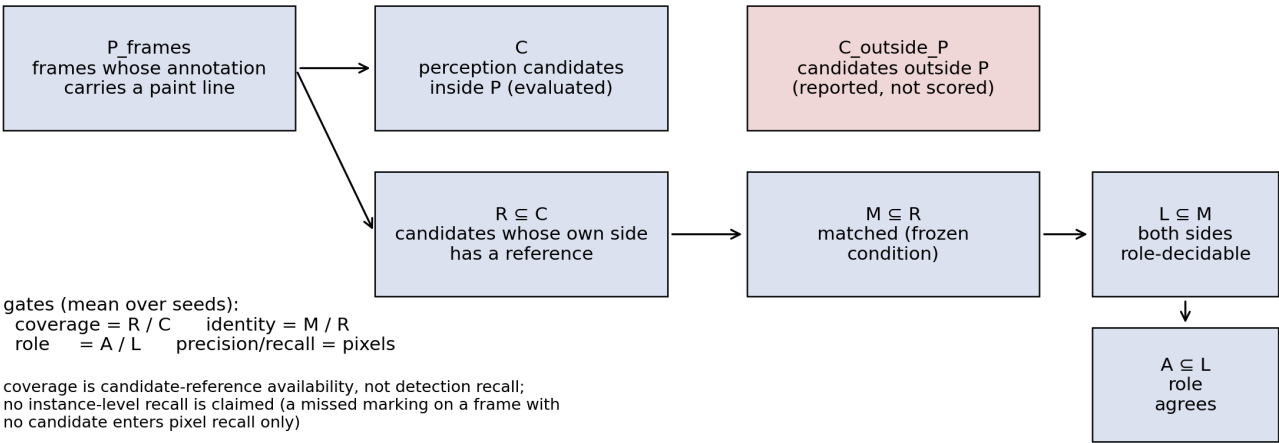
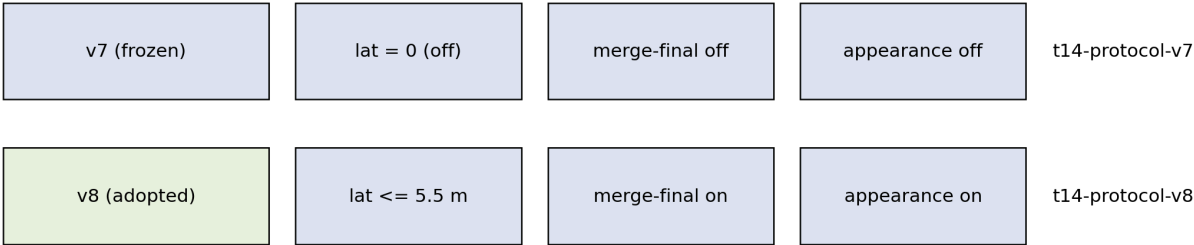


Fig. S18. Panel (b) of the main-text design and contract composite. Counting contract (schematic): every ratio has an explicit numerator and denominator, over the frozen counter definitions; coverage is candidate-reference availability, not detection recall.

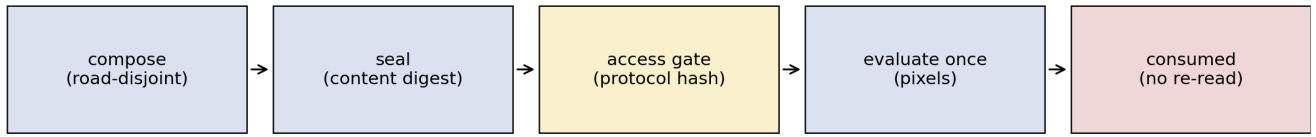
Protocol switch matrix (schematic) and the live hash



active version = t14-protocol-v8 protocol hash = f5ec944c5dd23b44
one constant switches all three changes; the hash changes with the version, so a seal/confirmation cannot be mislabelled

Fig. S19. Panel (c) of the main-text design and contract composite. Protocol switch matrix (schematic) and the live evaluation hash.

One-shot final confirmation flow (schematic) and the live seal



sealed frames = 200 digest = 798f84883eb75bde protocol = f5ec944c5dd23b44 confirmations = 1

a refused access is recorded but does not consume; a consumed set is a diagnostic set (freeze a new one)

Fig. S20. Panel (d) of the main-text design and contract composite. One-shot final-confirmation flow (schematic) with the live seal.

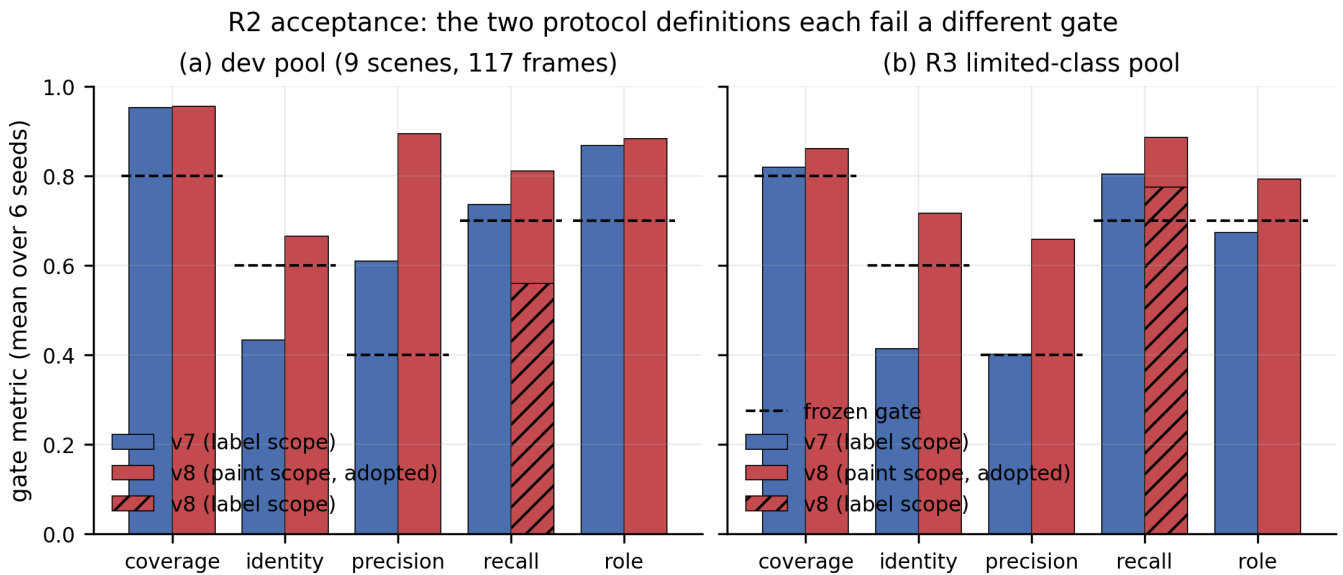


Fig. S21. Panel (a) of the main-text definition sensitivity composite. Gate metrics against frozen thresholds for the two protocol definitions, development pool (a) and limited-class pool (b).

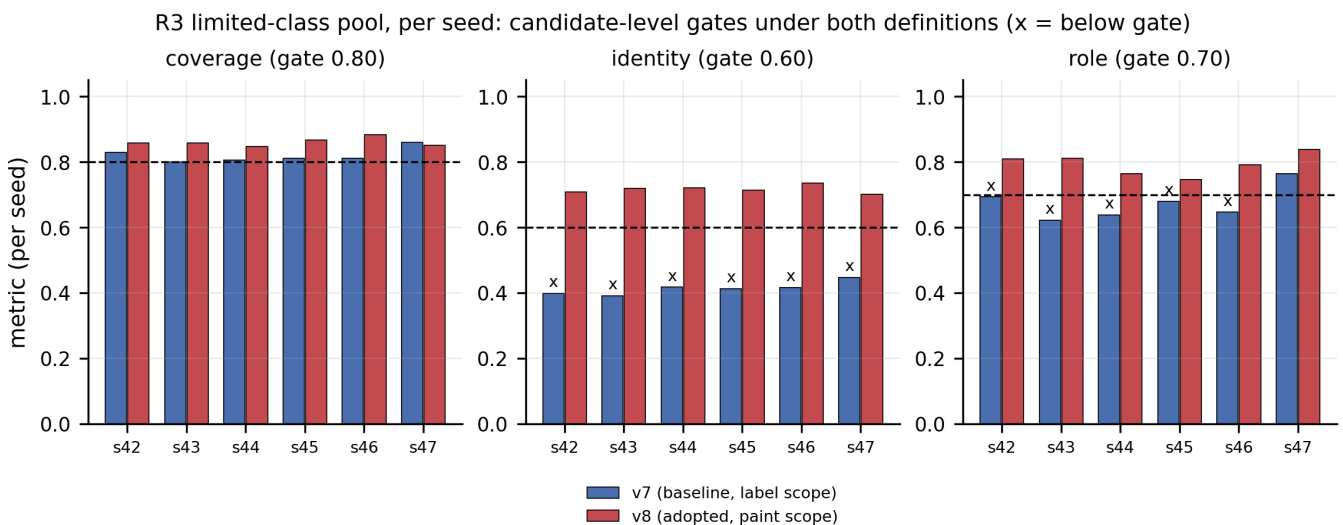


Fig. S22. Panel (b) of the main-text definition sensitivity composite. R3 limited-class pool, per seed: the three candidate-level gates under both definitions, drawn from the same verdict files as Table I (x = below gate).

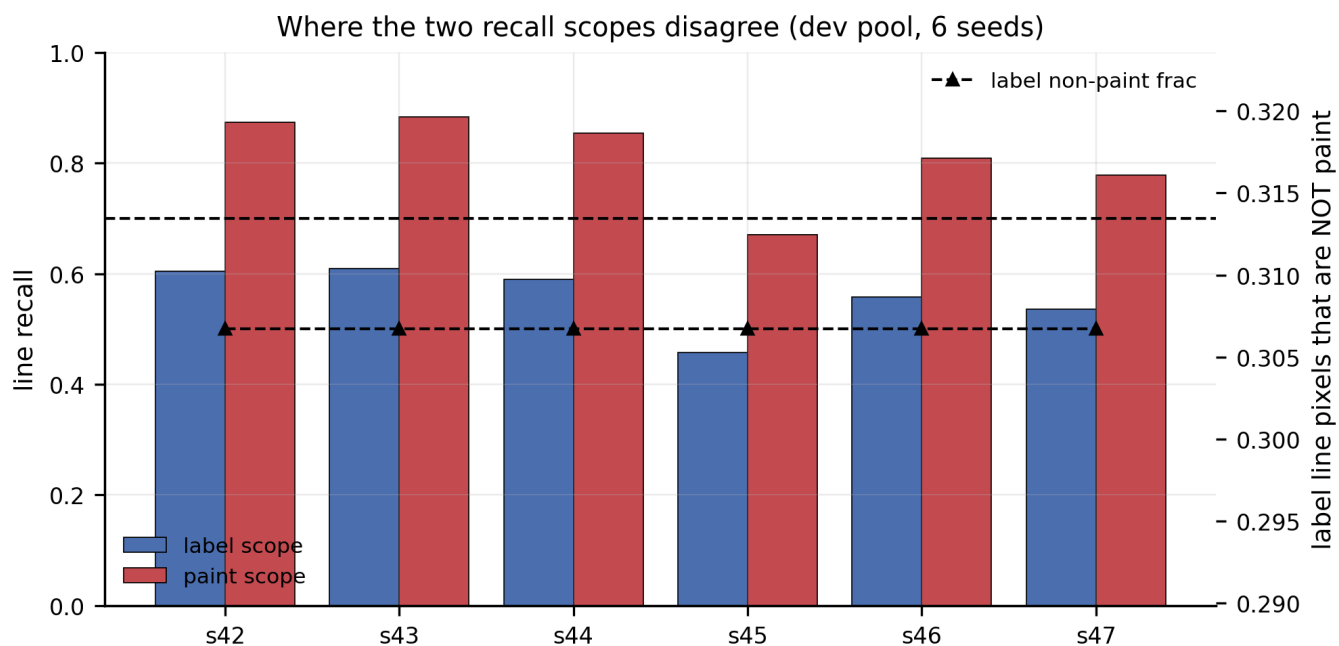


Fig. S23. Panel (c) of the main-text definition sensitivity composite. Per-seed label-scope versus paint-scope recall with the non-paint fraction of label line pixels (development pool).

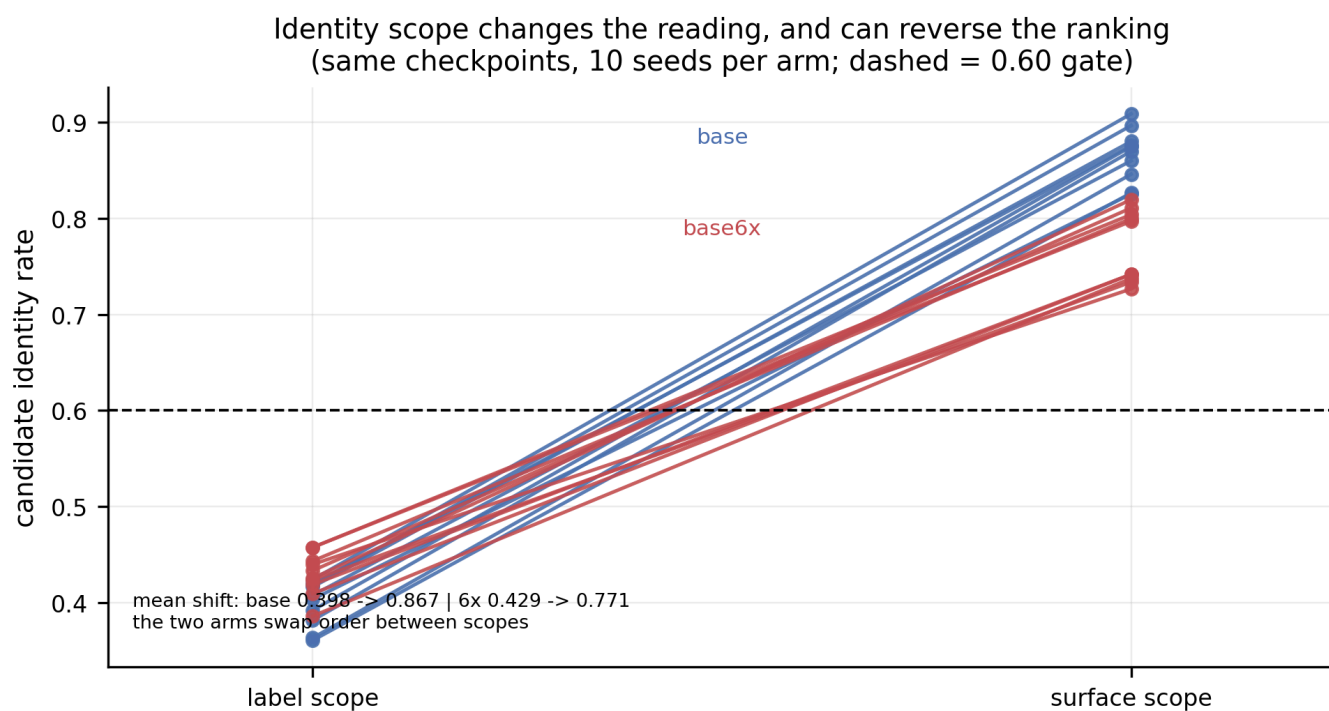
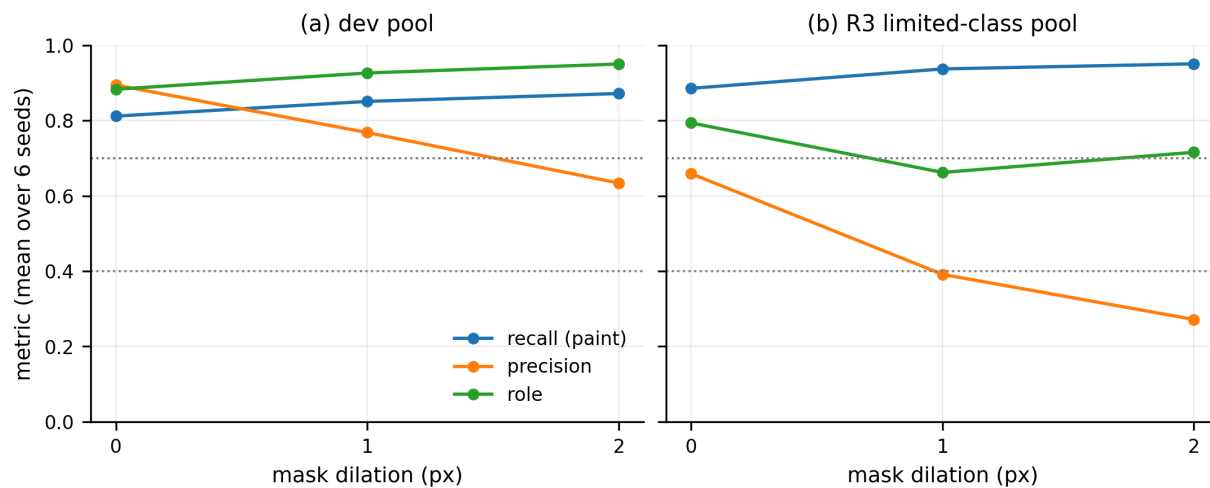


Fig. S24. Panel (d) of the main-text definition sensitivity composite. Identity scope changes the reading and can reverse the ranking: 20 checkpoints (two arms, ten seeds each) read in label and surface scope.

Boundary map: dilation buys recall and pays precision — R3 pays ~3x more, so no single value clears both pools



dev: the recall gate fails for 1/6 seeds at 0 px ($s45 = 0.671$) | R3: precision 0.39 and role 0.66 fall below their gates at 1 px

Fig. S25. Panel (a) of the main-text post-processing and composition composite. Boundary map: dilation buys recall and pays precision; the limited-class set pays about three times more.

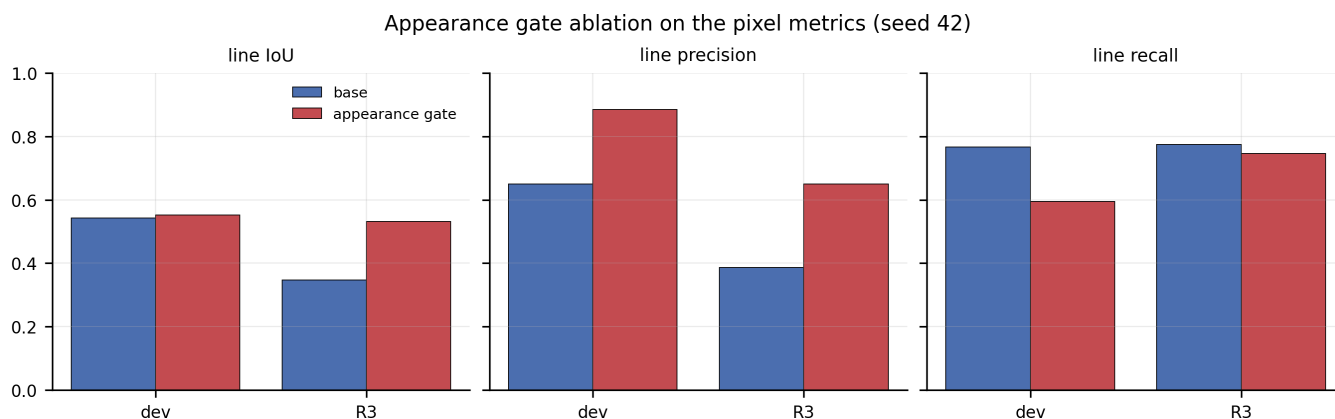


Fig. S26. Panel (b) of the main-text post-processing and composition composite. Appearance-gate ablation on the pixel metrics (line IoU, precision, recall) for both sets, seed 42.

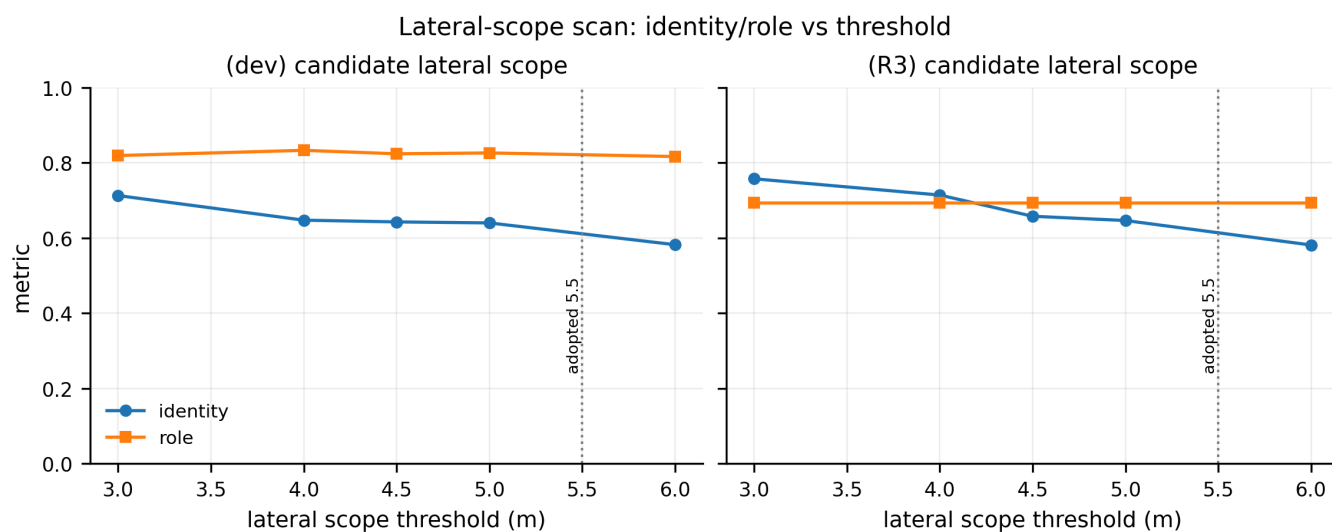


Fig. S27. Panel (c) of the main-text post-processing and composition composite. Lateral-scope scan on both sets: identity and role versus threshold, with the number of retained candidates.

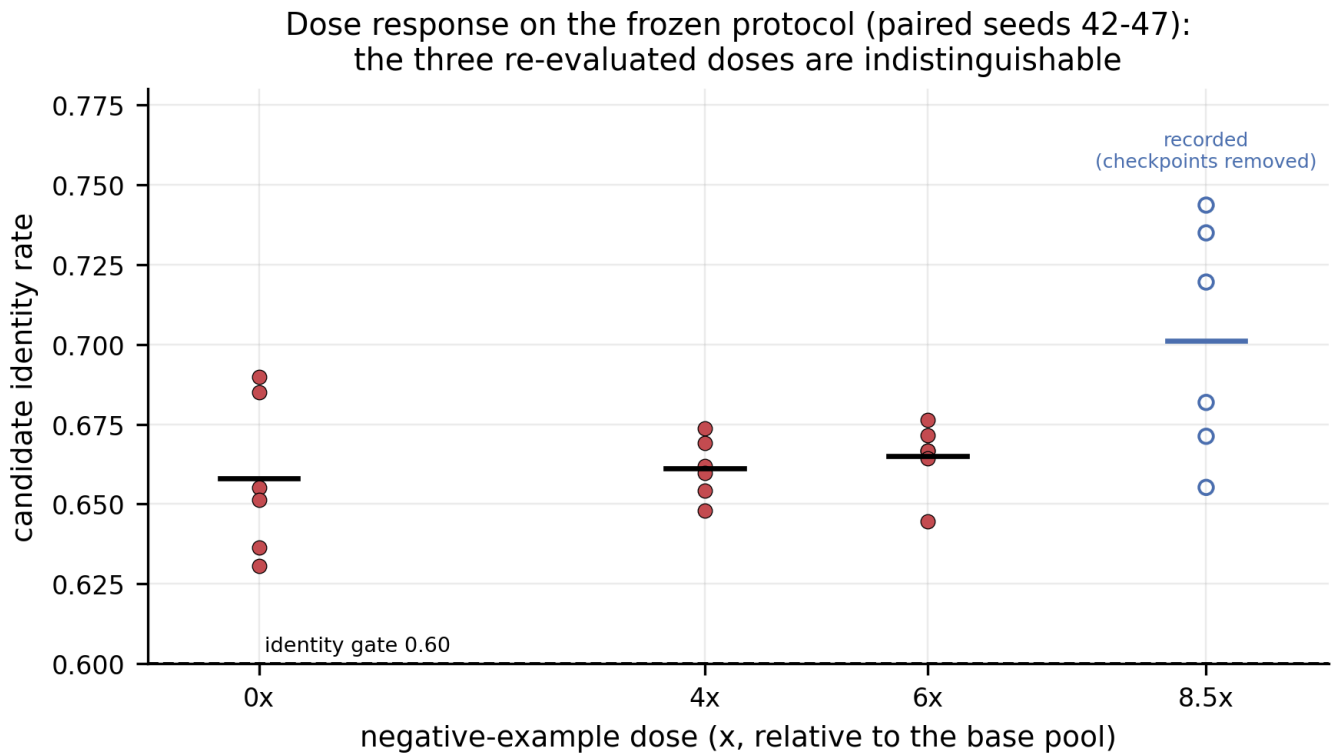


Fig. S28. Panel (d) of the main-text post-processing and composition composite. Dose response of the candidate identity rate (per seed and mean).

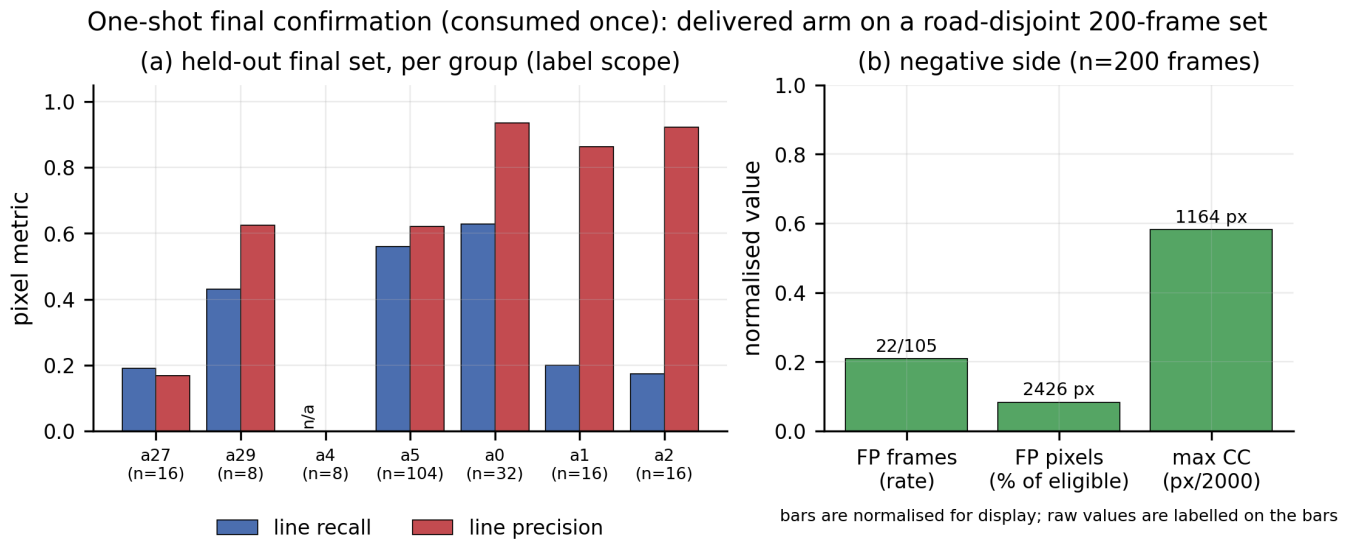


Fig. S29. Panel (a) of the main-text one-shot confirmation composite. One-shot final confirmation on the sealed road-disjoint set: per-group metrics (a) and negative-side diagnostics (b).

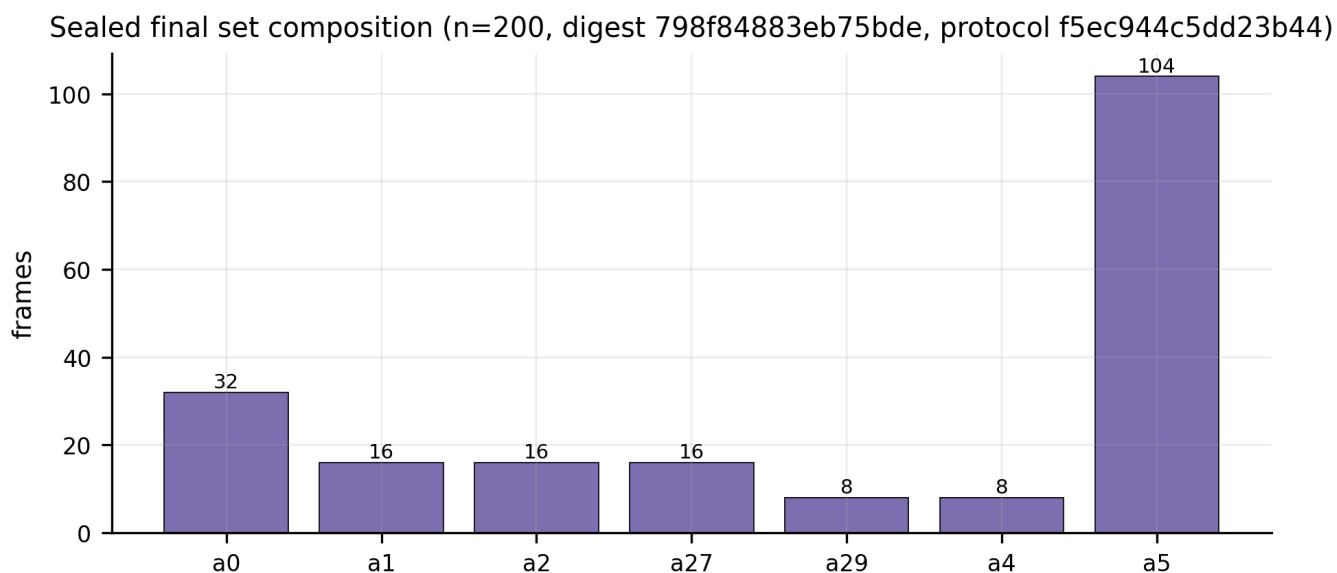


Fig. S30. Panel (b) of the main-text one-shot confirmation composite. Sealed final-set composition by group, with digest and protocol hash.

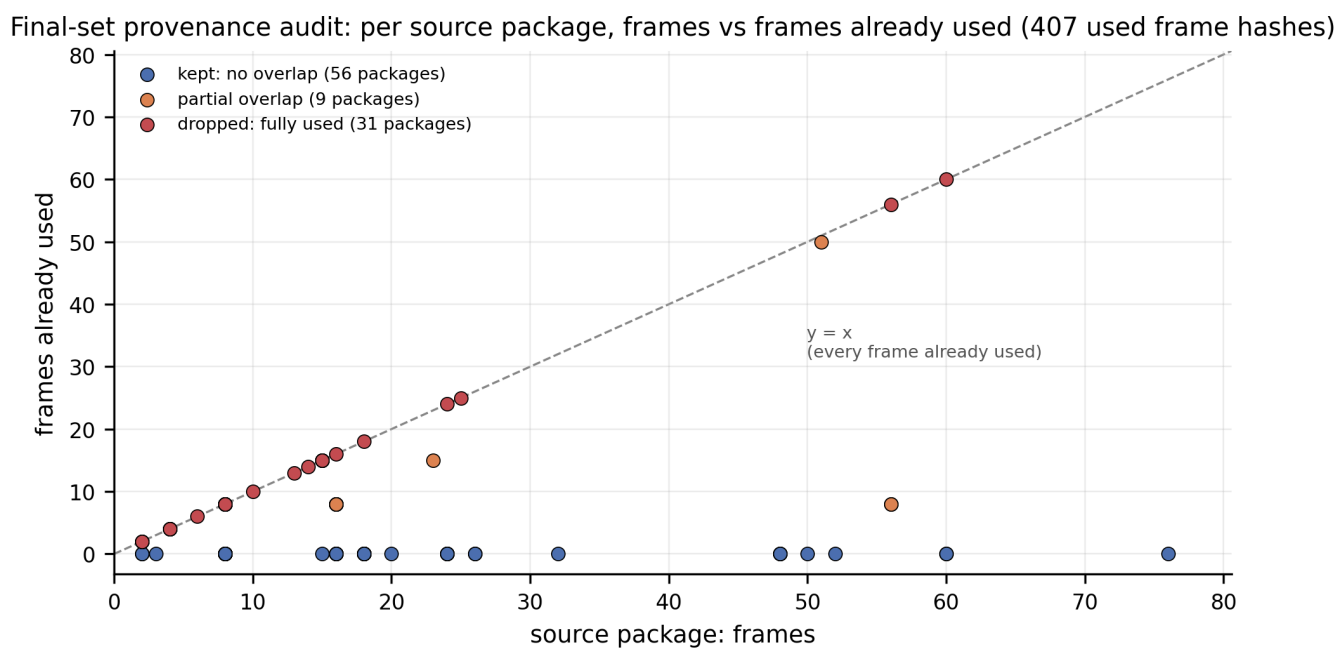


Fig. S31. Panel (c) of the main-text one-shot confirmation composite. Final-set provenance audit: content-level overlap of each source package with the used frame set.



Fig. S32. Panel (a) of the main-text closed loop composite. Closed-loop trade-off: centre crossings versus sensor lane-source rate, four runs per arm.

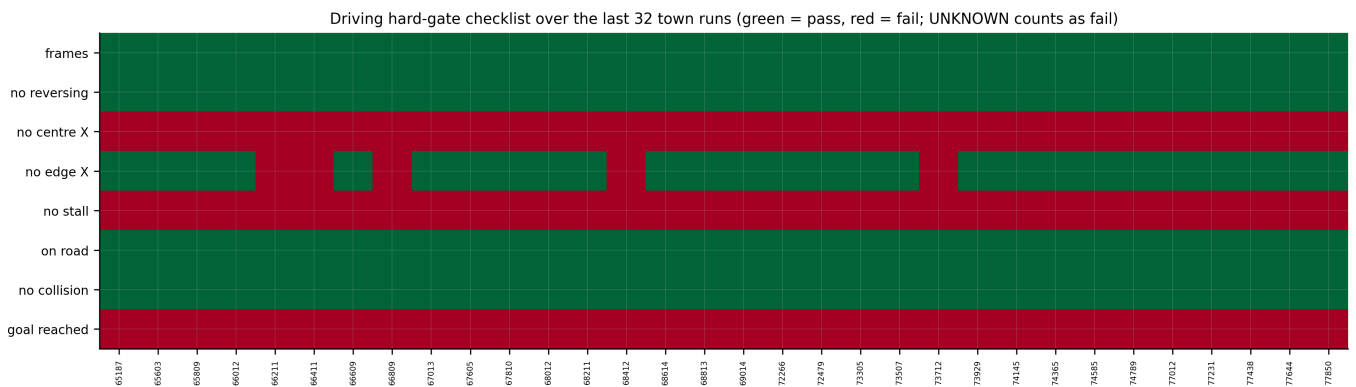


Fig. S33. Panel (b) of the main-text closed loop composite. Driving hard-target checklist over the most recent town runs (green = pass, red = fail; unmeasured target counts as fail).

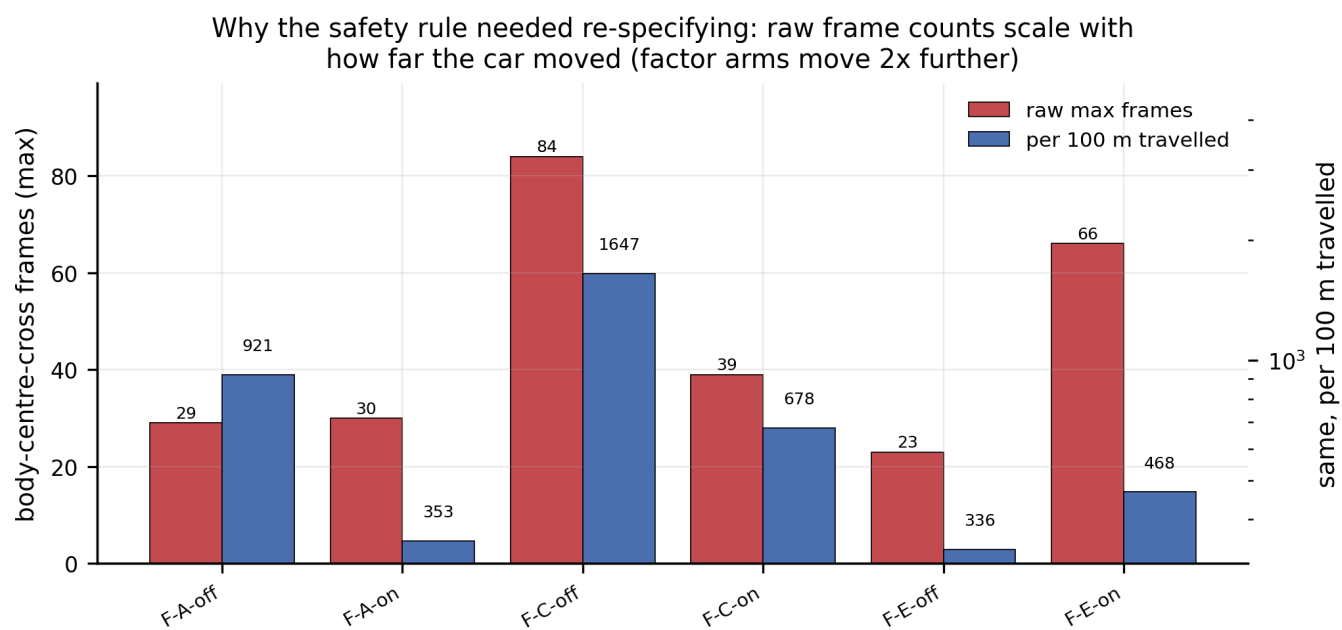


Fig. S34. Panel (c) of the main-text closed loop composite. Raw crossing counts versus the same quantity normalised per 100 m travelled (log scale).

Safety arbitration ladder (schematic, from ARBITRATION_RULES)

stale_sensor_planner	degraded
path_hold	degraded
no_drivable_path	minimal_risk
perception_lane_unavailable	minimal_risk
road_surface	minimal_risk
body_off_pavement	minimal_risk
path_blocked	minimal_risk
scattered_obstacle	degraded
path_grazes	degraded
lane_boundary_recovery	degraded
planned_boundary_crossing	degraded
stationary_recentre	degraded
body_crosses_boundary	minimal_risk
path_off_lane	minimal_risk
path_near_lane_edge	degraded
obstacle_very_close	degraded
soft rules accumulate caps; a minimal_risk rule stops the tick (order is the evaluated order, not severity) obstacle_risk	minimal_risk

Fig. S35. Panel (a) of the main-text deadlock mechanism composite. Safety arbitration ladder (schematic, from the monitor's rule registry).

Deadlock anatomy: a parked, off-centre car is refused every path (12 town runs, current body inside in all frames)

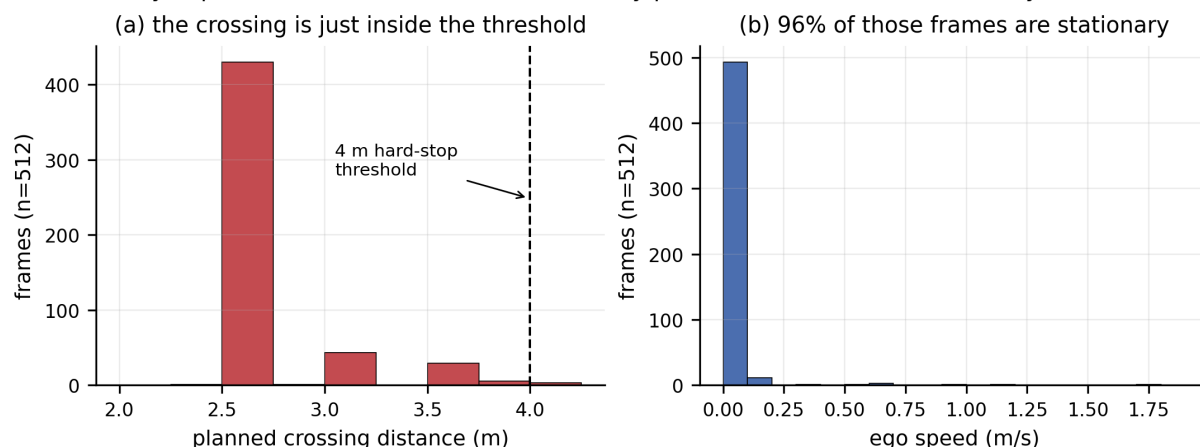


Fig. S36. Panel (b) of the main-text deadlock mechanism composite. Deadlock anatomy: the planned crossing sits just inside the 4 m threshold (a) and the car is stationary in 96% of those frames (b).

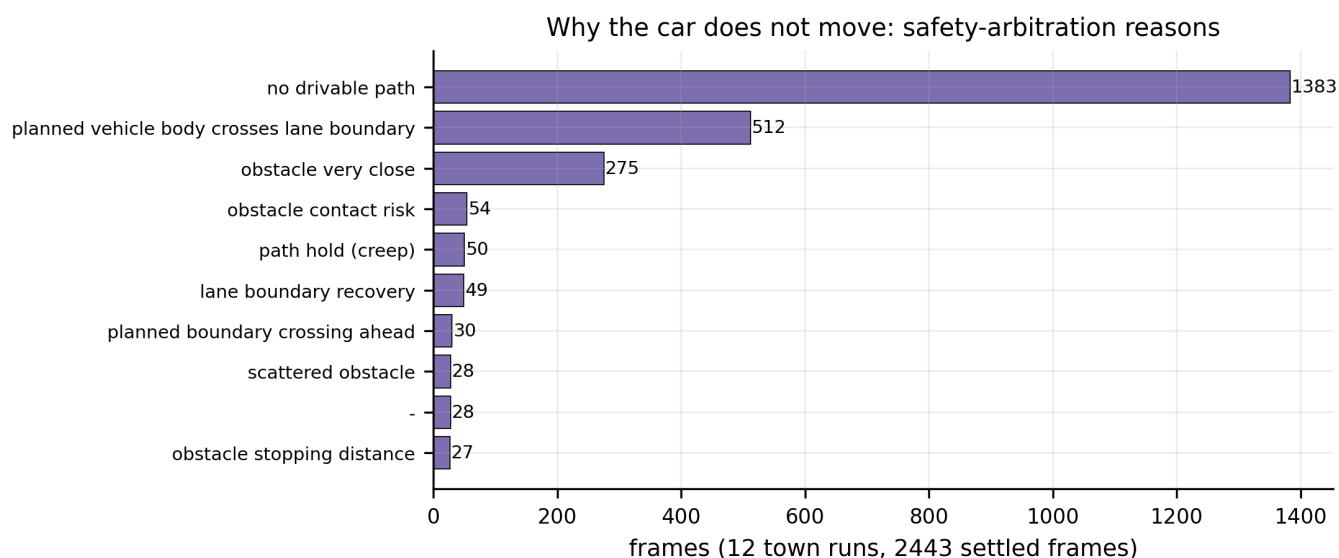


Fig. S37. Panel (c) of the main-text deadlock mechanism composite. Why the car does not move: safety-arbitration reasons over 2443 settled frames.

Lane-acceptance evidence: the 'on-pavement' gate is driven by the BEV obstacle layer, not by 'off the pavement'

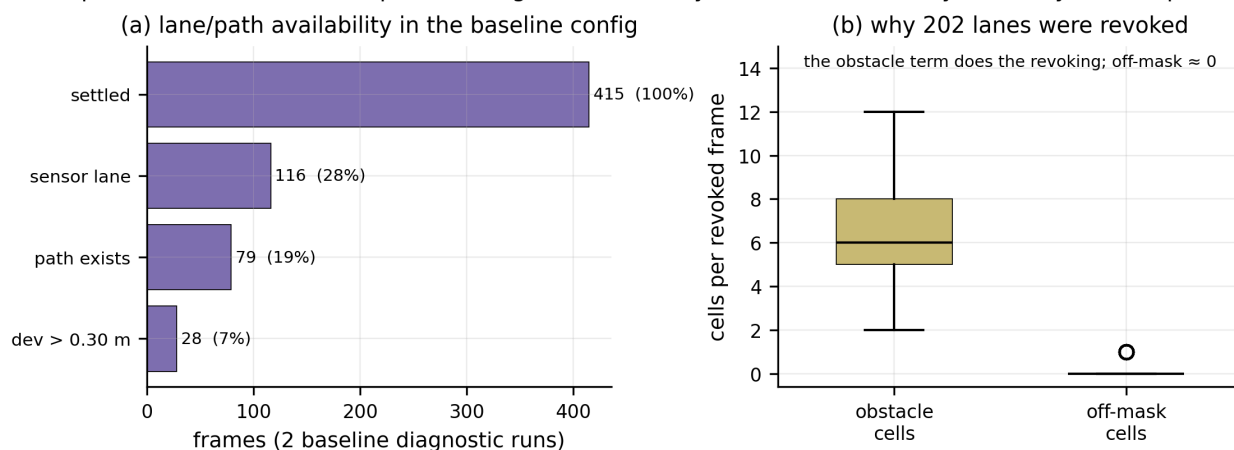


Fig. S38. Panel (d) of the main-text deadlock mechanism composite. Lane-acceptance evidence: baseline availability funnel (a) and the evidence behind the revoked lanes (b).

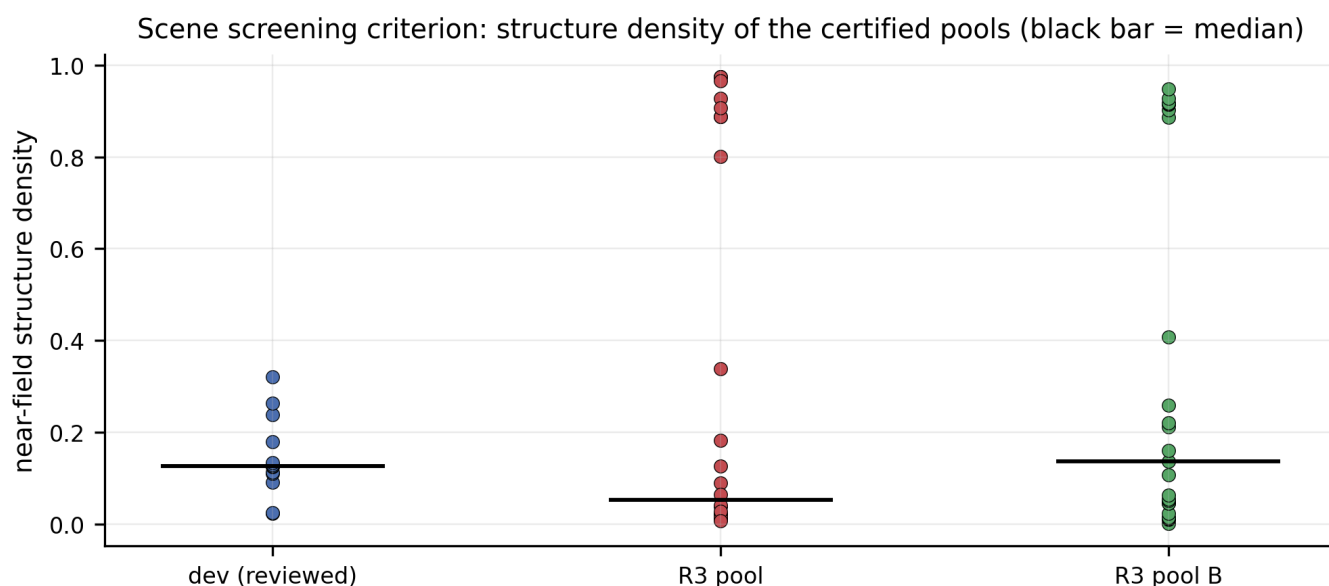


Fig. S39. Panel (a) of the main-text rigour and audits composite. Scene screening criterion: near-field structure density of the certified pools (black bar = median).

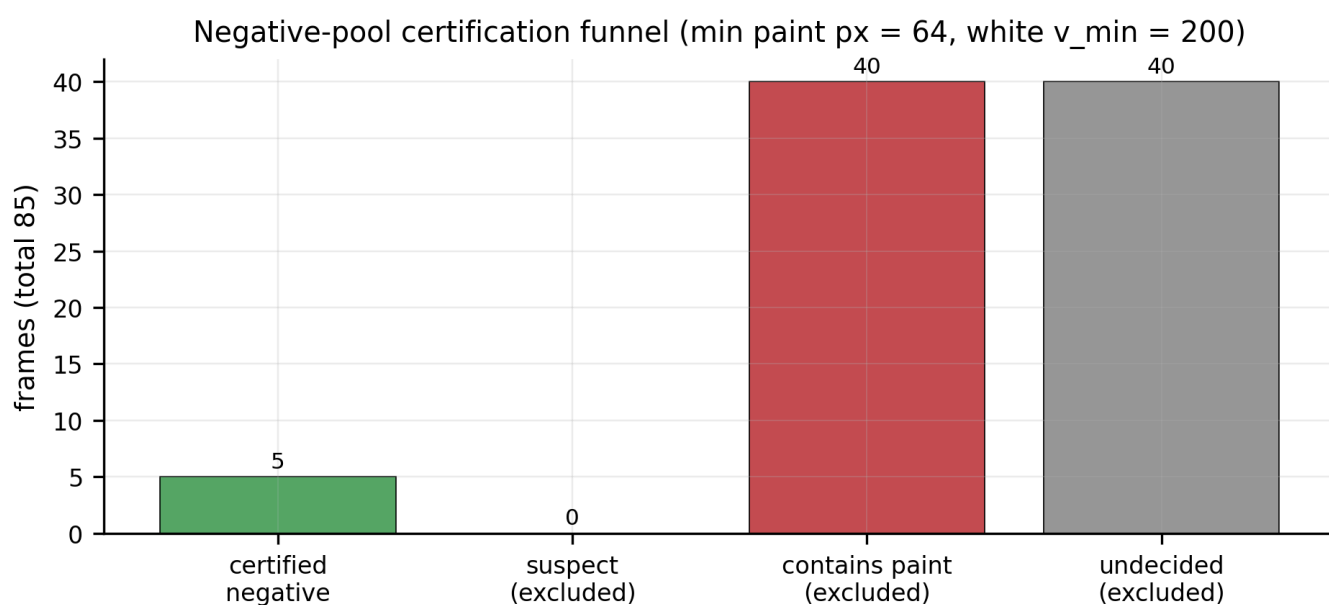


Fig. S40. Panel (b) of the main-text rigour and audits composite. Negative-pool certification funnel: frames admitted as certified negatives versus excluded categories.

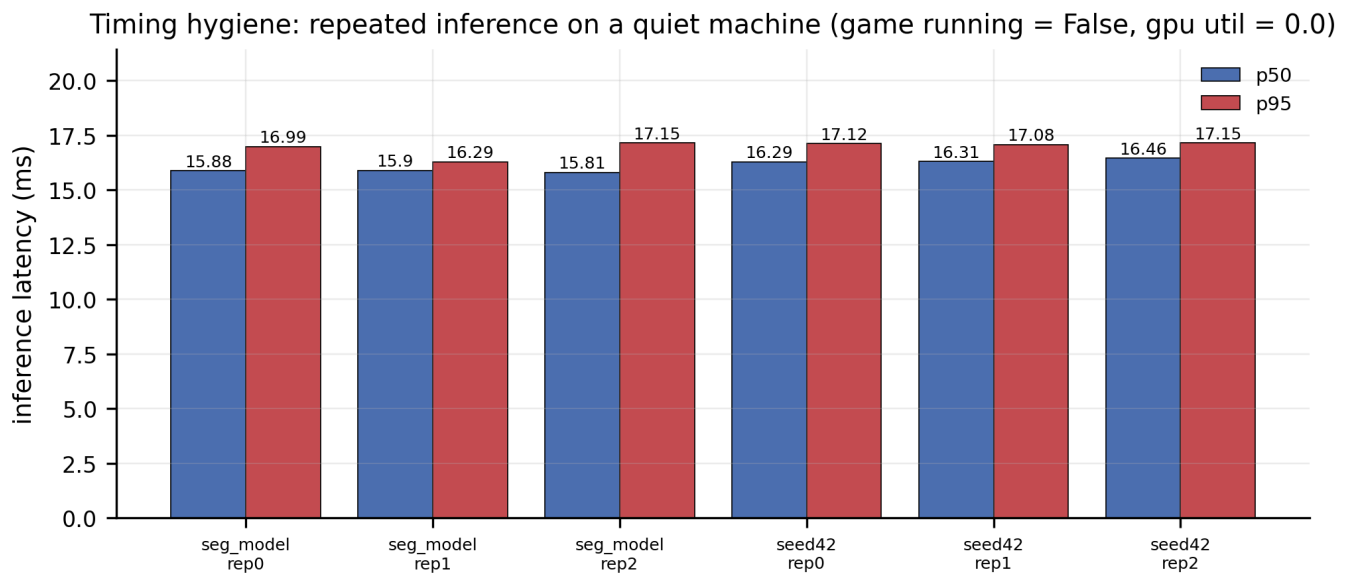


Fig. S41. Panel (c) of the main-text rigour and audits composite. Timing hygiene: repeated inference measurements on a quiet machine.

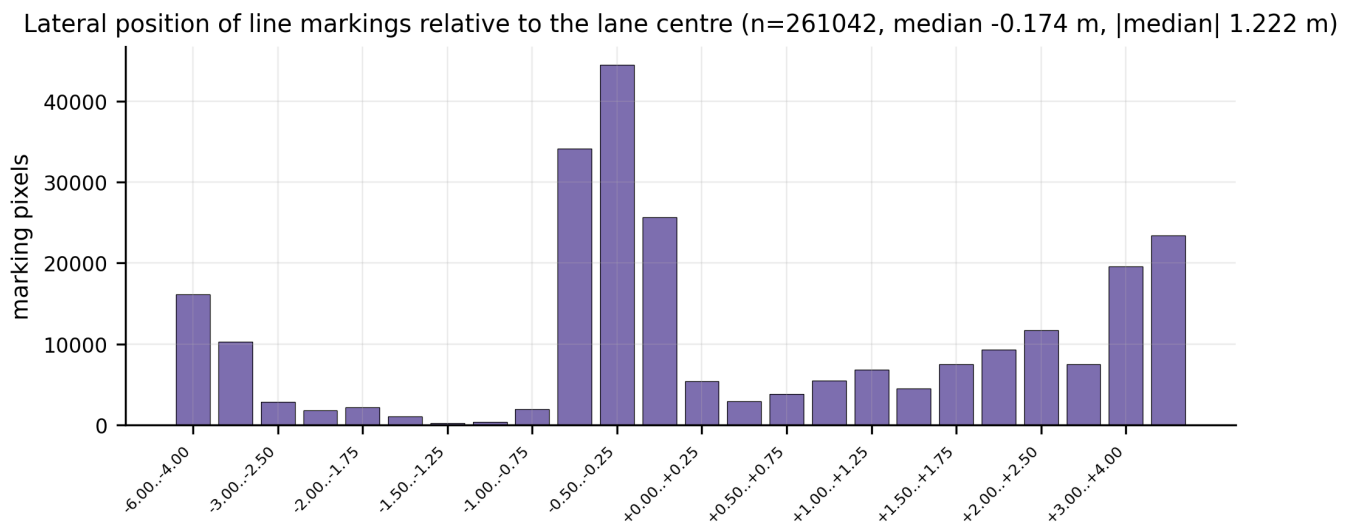


Fig. S42. Panel (d) of the main-text rigour and audits composite. Lateral position of line markings relative to the lane centre, pooled over the human-revised development pool.